

# Memory-Sovereign Inference: Output-Exact Execution Beyond Full Residency

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Technical report

**Abstract:** Storage-backed inference is easy to demonstrate and easy to overclaim. A low process resident set does not bound charged page cache, a process-local device reading does not establish whole-board use, and successful generation does not show that asynchronously reused buffers held the intended bytes. We present a reference architecture and falsifiable execution-evidence certificate. A certificate records an immutable representation, separates semantic workload demand from scheduler requests and cumulative traffic, names a per-tier resource-authority vector and exactness horizon, and records implementation-independent reuse invariants with the transitions actually tested.

At one Qwen3-Next system identity, the complete native tensor representation is 45.08 GiB. The stock router selects every one of the 48×512 managed layer-expert objects during the 32K pre-fill. Their duplicate-free, overlap-free canonical-extent union yields a conservative 43.59375 GiB semantic-demand lower bound. Both quantities exceed the declared 34 GiB full-residency envelope; the demand lower bound also exceeds the 11 GiB host-hard plus physical 24 GiB-device envelope by 8.59375 GiB<sup>C1</sup>. The LRU64 execution remains within its host-hard/GPU-audited resource contract. Against one prespecified zero-cache oracle, all 64 token IDs, every byte of all 64 complete 151,936-float logit rows, all 3,408 recorded route events, response bytes, and the seven-field consumer and two-entry destination identities are exact. We call this relation full-output-horizon exactness; recurrent-state and upstream-runtime equality are not claimed.

In a matched source campaign, buffered P completes exactly but reaches the 11 GiB host ceiling and records 33,481 memory.max events. Blocking one-window direct D and the complete eight-window asynchronous F component are exact with positive margin and zero limit events. Across six counterbalanced pairs, F completes in 32.3% of D's wall time ( $3.10\times$  reciprocal); all F/D ratios lie in [0.3211, 0.3283], at identical physical source bytes

per output<sup>C2</sup>. The contrast jointly changes queue depth, overlap, and lifecycle implementation. In a separate package of later fault-instrumented executables, all fourteen prespecified deterministic control/fault cells satisfy their validators across Qwen3-Next and Gemma 4 adapters<sup>C3</sup>. They support fail-closed behavior for named transitions in those binaries, not a reliability rate, race-freedom claim, or fault attribution to the earlier 32K+64 executable. The principal experiment remains one fixed model, workload, runtime, device, and 64-output horizon.

## 1 Introduction

CPU offload, memory mapping, SSD streaming, expert caching, and speculative generation all trade capacity for traffic or latency [16, 2, 5, 1, 21, 6, 23]. The difficult question is what a successful run proves. A process RSS limit does not bound the page cache. A device allocation reported by one process does not include the rest of the board. A cache hit is invalid if a slot is recycled before its last consumer. A final token match can hide stale reads that did not affect one prompt. Consequently, a model can appear to fit an operator's memory budget without establishing either resource admissibility or execution integrity.

We call the stronger property memory sovereignty. A memory-sovereignty evidence record states what bytes exist and are semantically demanded, which authority governs each resource tier, what equivalence relation holds for which horizon, and which reuse invariants were tested. A claim class then states what that record qualifies. Host enforcement and GPU accounting need not use the same mechanism: here, process-tree host memory is hard-limited with cgroup v2, whereas GPU use is declared and audited from whole-board observations; the physical board supplies a separate hard capacity ceiling. The authority vector is part of the result, not a presentation detail.

The architectural observation is that four concerns

commonly fused into a tensor allocation should be separated: canonical storage ownership, physical residency, stable execution addresses, and scheduling. The architecture realizes this separation through a shared Plan. The plan maps logical objects to byte-preserving source extents, fixed host windows, reusable GPU slots, stable addresses, and generations. Model adapters name the objects required by a layer; the native graph, kernels, router, KV state, and sampler remain authoritative.

The contribution is an execution-certification methodology and reference realization for storage-backed inference, not a new offload primitive. Direct I/O, bounded windows, stable addresses, generations, caching, and prefetch appear across FlexInfer, MESH, Palm, MoE-Direct, FluxMoE, and DeepNVMe [6, 12, 18, 19, 13, 5]. The differentiated result is a publication evidence package: the principal Qwen3-Next experiment executes a workload whose semantically demanded managed weights exceed the declared and host-hard/ device-physical envelopes with full-output-horizon exactness and hard complete- process-tree host accounting; a later matched-source experiment shows direct ownership changing the prespecified admission outcome and the complete asynchronous component completing in 32.3% of a blocking direct reader’s wall time ( $3.10\times$  reciprocal); and separate fault-instrumented binaries pass named fail-closed transitions across two adapters. The package does not collapse these executable identities into one certificate.

This report makes four contributions:

- It demonstrates full-output-horizon exact native Qwen3-Next execution under  $E_{64-out}$  when a 43.59375 GiB semantic-demand lower bound exceeds a declared 34 GiB full-residency envelope, with hard complete-process-tree cgroup-charge accounting and audited whole-board GPU margin<sup>C1</sup>.
- Relative to the tested buffered P path, it shows that direct source ownership changes the prespecified hard-margin admission outcome, and it shows the complete eight-window asynchronous component finishing in 32.3% of a blocking one-window direct reader’s wall time ( $3.10\times$  reciprocal) at identical physical source bytes<sup>C2</sup>.
- In later fault-instrumented builds, it tests all fourteen prespecified control/fault cells through two real model adapters, supporting named fail-closed transitions at those tested identities<sup>C3</sup>.
- It defines a falsifiable, implementation-independent certificate and a shared cross-adapter Plan realization, then records the temporal, Qwen-specific, and Gemma experiments without pooling their identities.

The report also records stopped lines. A GPT-OSS packing optimization did not repair its physical carrier, and a Llama-3.3 placement result passed resources but

failed its task-admission criterion. These negatives prevent local engineering wins from being promoted into an inference-system claim.

## 2 The Memory-Sovereignty Contract

### 2.1 Certificate Tuple and Authority Grades

We represent an execution-evidence record as  $\mathcal{C} = (I, B, A, E, L)$ .  $I$  is an immutable inventory and provenance record;  $B$  is the boundary vector by memory tier;  $A$  assigns each boundary an authority grade;  $E$  names the equivalence relation, oracle identity, and horizon; and  $L$  names the reuse invariants and transition set actually tested. Authority grades distinguish a hard-enforced limit, a physical capacity ceiling, an instrumented high-water mark, and a sampled audit. A certificate may therefore report host=hard and GPU=audited without treating the two claims as equivalent.

The tuple is a record schema, not by itself a pass criterion. We attach compositional claim classes:  $R$  states that the complete representation cannot be resident under  $B$ ;  $W$  states the stronger workload result that a semantic-demand lower bound cannot be resident under  $B$ ;  $E[E, h]$  names an exact relation and horizon; and  $L[F]$  names a tested transition set. A memory-sovereign execution claim requires  $R$  or  $W$ , a passing resource predicate under the disclosed authority vector, a named  $E$ , and clean applicable lifecycle closure at one execution identity. Additional fault cells may form a separate  $L$  record, but do not transfer between binaries implicitly. Classes are compared dimension by dimension; the paper defines no scalar grade that would conflate hard enforcement, physical capacity, and sampled audit.

The inventory separates four quantities.  $T_{repr}$  is the complete canonical tensor payload under a stated manifest inclusion rule.  $U_{sem}^{LB}$  is a conservative lower bound on unique logical canonical bytes selected by the authoritative model graph and consumed at least once.  $U_{sched}$  is the deduplicated union requested by a source or cache policy.  $Q_{run}$  is cumulative logical or physical traffic, including rereads and alignment padding. Scheduler activity and traffic are never substituted for semantic demand.

Let  $H_{cap}$  be the enforced process-tree host allowance,  $G_{decl}$  the declared and audited GPU allowance,  $G_{phys}$  the physical board capacity, and  $R_{policy}$  an operator-declared reserve for non-target host state. Let  $S$  denote the canonical bytes managed from storage. The evaluated predicates are:

$$S > H_{\text{cap}} - R_{\text{policy}}, \quad (1)$$

$$T_{\text{repr}} > H_{\text{cap}} + G_{\text{decl}}, \quad (2)$$

$$U_{\text{sem}}^{\text{LB}} > H_{\text{cap}} + G_{\text{phys}}. \quad (3)$$

Equation 1 is policy-relative:  $S$  cannot coexist in the host allowance while preserving the declared reserve. Equation 2 makes complete representation residency infeasible within the declared placement envelope. Equation 3 is a semantic-workload result against a host-hard/device-physical envelope. Adding host and GPU bytes is a relaxation despite their operational non-fungibility; exceeding the sum is therefore sufficient for nonresidency.

Separately, the evaluated workstation exposes 16,667,594,752 host bytes through Linux and a physical 25,769,803,776-byte GPU board. The Qwen3-Next complete representation exceeds their sum by 5.56 GiB, and its demanded-expert lower bound exceeds it by 4.07 GiB <sup>C1</sup>. This is an operating-system-visible capacity observation, not a claim about all physically installed or firmware-reserved host DRAM.

A virtual mapping does not discharge these predicates. Buffered I/O can move model bytes into charged page cache while process RSS stays low. Claim-bearing runs therefore place the complete process tree in a cgroup v2 boundary, disable swap, and record memory.current, memory.peak, swap, OOM, and memory.max events. Device observers sample whole-board allocation and compute-process ownership. GPU compliance is audited rather than hard-partitioned: a valid cell has positive margin below its declared allowance and no foreign compute process; the board size is used separately as a physical ceiling.

We call the host predicate hard complete-process-tree cgroup-charge accounting: it covers all memory charges that cgroup v2 attributes to the target hierarchy, including charged anonymous memory, file cache, and kernel memory. It is not a total-causal-host-byte claim. Shared pages retain Linux charge semantics; firmware, device-internal buffers, unaccounted driver allocations, and processes outside the hierarchy are outside this predicate. No cooperating process outside the hierarchy participates in the recorded cells. The principal telemetry archives the aggregate high-water mark and events, not a per-class memory.stat decomposition.

## 2.2 Exactness Is Scoped by Identity and Horizon

For the principal experiment, the named relation is

$$E_{64\text{-out}} = E_{\text{logits}} \wedge E_{\text{tokens}} \wedge E_{\text{response}} \wedge E_{\text{routes}} \wedge E_{\text{consumer}} \wedge E_{\text{destination}}. \quad (4)$$

Table 1: Implementation-independent certificate obligations. Generations, epochs, and consumer counts are this prototype’s realization, not universal requirements.

| Field  | Required evidence                                                                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $I$    | Representation, semantic-demand and scheduler unions, cumulative traffic, inclusion rules, and immutable provenance.                                                                   |
| $B, A$ | Per-tier allowance and authority grade; process-tree/page-cache and whole-board attribution; margins and event counters.                                                               |
| $E$    | Same-build oracle, compared surfaces, and exact horizon; semantic metrics remain separate.                                                                                             |
| $L$    | No publication before movement; requested object/version identity; no reuse before consumers finish; reset/shutdown drain or invalidate; invalid completions cannot make data visible. |

Here  $E_{\text{logits}}$  covers every byte of all 64 complete logit rows; the remaining terms cover the corresponding ordered tokens, response bytes, recorded routes, consumer identity, and destination identity. We use full-output-horizon exactness under  $E_{64\text{-out}}$ ; the phrase does not imply equality of every internal state.

The consumer term compares a seven-field record naming the base and adapter commits, graph, router, expert kernel, reduction, and scheduling boundary. The destination term compares the two-entry model-memory placement record. Both are compared structurally, rather than inferred from output equality.

The strongest oracle compares complete logits and generated token IDs against the same build, graph, tensor representation, prompt, and deterministic sampler. Sparse routes and recurrent state are included when exposed by the runtime. The term exact applies only to the recorded scope. A short test does not certify a longer execution, and a prefix comparison does not certify the remaining outputs. Natural-task F1 is separate evidence: it measures user-visible utility but cannot establish internal equality. The principal experiment’s LRU64-versus-zero-cache comparison tests cache-policy equality within its recorded executable. A separate zero-cache source-path canary compares an eight-window direct source against its synchronous-source control on a 256-token prompt and eight-output horizon. It compares all eight complete logit rows, tokens, response bytes, 384 ordered route events, consumer and destination identities, and clean closure. Its one prefill event at each of 48 layers selects all 512 experts, covering all 24,576 managed layer-expert objects. This is an eight-output source-path result, not part of the 64-output cache-policy horizon. Equality to an unmodified upstream runtime and di-

rect recurrent/hybrid-state bytes are unavailable and not claimed.

This distinction matters for sparse and hybrid models. Router decisions amplify numeric differences into future I/O; recurrent state differences can alter later speculative acceptance. The implementation retains native quantized bytes and the existing graph so that a source-path comparison is not also a model-conversion experiment.

### 2.3 Lifetime Is Part of Correctness

A reusable resource version must not be published before its movement completes, must denote the requested object and version, and must not be reused before declared consumers finish. Reset and shutdown must drain or invalidate outstanding work, and an invalid completion must not make data visible. These properties do not prescribe a bookkeeping mechanism. The implementation realizes them with request epochs, slot generations, readiness events, tickets, and consumer counts. The reported evidence covers tested implementation behaviors, not a formal proof of race freedom.

## 3 Design

Figure 1 separates the data and control paths. Canonical tensor extents flow through fixed host windows into reusable GPU slots. The control path resolves a model-layer request to an object bundle, initiates bounded reads and copies, and publishes stable execution addresses after the tested identity and readiness checks. External observers account for the resource envelope independently of the model process.

### 3.1 A Shared Cross-Adapter Plan Core

The Plan comprises an immutable manifest and mutable execution state. The manifest records object identifiers, logical sizes, source extents, alignment and padding, compatible windows and slots, and stable device addresses. Object identifiers express model semantics rather than file layout. An adapter can therefore combine several extents into one logical object without exposing the container layout to the graph.

Mutable state associates each window and slot with an object, generation, request epoch, readiness event, and live-consumer count. Replacement policy is separate. Demand loading, LRU, prediction, and pinning select different future objects but use the same lifecycle interface. Transfer between Qwen and Gemma changed object mapping and adapter code; it did not establish a general abstraction theorem.

Table 2: Shared-core and adapter responsibilities. Cross-adapter reuse does not imply unrestricted model neutrality.

| Shared Plan core               | Model/runtime adapter             |
|--------------------------------|-----------------------------------|
| Extents and slot compatibility | Container object discovery        |
| Versioned resource state       | Bundle and layer-demand semantics |
| Readiness and publication      | Graph and kernel integration      |
| Consumer and recycle lifetime  | Router, KV, and recurrent state   |
| Reset/shutdown invalidation    | Model-specific oracle surfaces    |
| Telemetry and validators       | Native layout interpretation      |

### 3.2 Bounded Source Service

The source service opens canonical files for direct I/O and preallocates a fixed set of aligned host windows. It computes covering reads, submits no more than the configured in-flight limit, and records logical bytes, physical bytes, padding, completion status, and latency. Short reads, negative completions, extent overflow, and object-size disagreement are rejection conditions in the tested implementation. Padding is excluded from logical tensor bytes.

Each completion carries the object, extent, window, request epoch, and generation present at submission. A completion with a mismatch does not publish host readiness in covered unit tests. After all extents are assembled, the window becomes eligible for H2D. Fixed windows give the explicit bound

$$M_{\text{source}} \leq N_w W + M_{\text{queue}} + M_{\text{meta}}, \quad (5)$$

where  $N_w$  is the number of windows and  $W$  their aligned size. Direct I/O avoids a second copy of the streamed payload accumulating in the page cache; complete cgroup accounting remains necessary for every other host allocation.

### 3.3 Stable Slots and Bundle Publication

Model runtimes often capture tensor addresses in graph nodes or device descriptors. The implementation reserves a bounded slot pool whose addresses remain stable while logical identity and generation change. Before a layer executes, an adapter requests its object bundle. In the tested implementation, all components are checked and reserved before one publication state transition makes the bundle visible. Unit negative controls and real-adapter deterministic cells cover short and failed reads, partial bundles, out-of-order completion, stale generations, wrong objects, held consumers, reset, and shutdown. This is not described as transactional: untested schedules and races remain open.

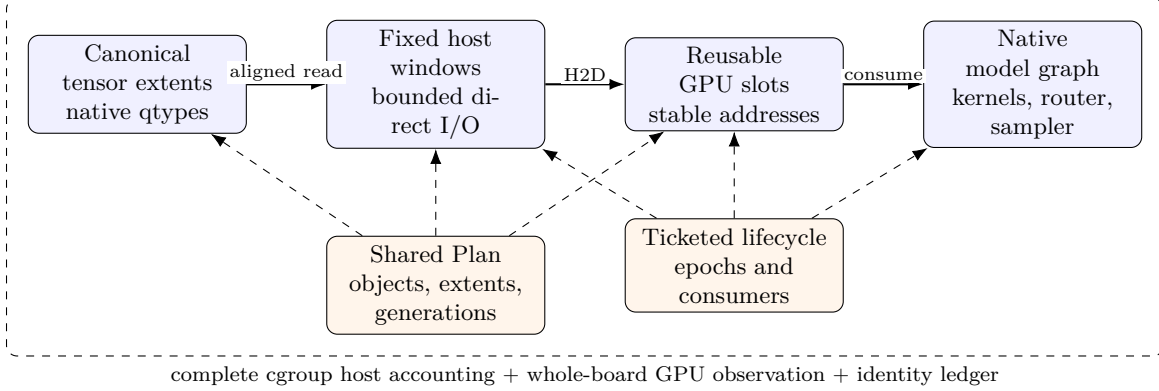


Figure 1: The architecture separates canonical ownership, bounded physical residency, stable execution addresses, and scheduling. The model graph consumes native tensors at stable slots; the Plan and ticket protocol determine which bytes those addresses denote.

### 3.4 Separate Window and GPU-Slot Lifecycles

Host windows and GPU slots are different resources. Their state sequences are therefore specified separately:

window: Free  $\rightarrow$  Reading  $\rightarrow$  HostReady  
 $\rightarrow$  Copying  $\rightarrow$  Free,

GPU slot: Vacant  $\rightarrow$  Loading  $\rightarrow$  DeviceReady  
 $\rightarrow$  InUse  $\rightarrow$  Recyclable  $\rightarrow$  Vacant.

A host window can return to Free after its H2D dependency completes; a GPU slot cannot become Recyclable until the copy completes and its generation has zero live consumers. Tickets carry object, generation, epoch, and resource identity across the two state machines. Reset advances the epoch after the tested outstanding-work checks; shutdown also checks submitted source operations. Covered invalid transitions become explicit errors. The tested implementation fails closed for the named deterministic cells in Section 5.4; no claim extends to all integrated faults.

### 3.5 Temporal and Spatial Amortization

The same service boundary supports two distinct economies. Spatial amortization retains recently used objects so future traversals avoid storage and H2D traffic. Temporal amortization commits several output tokens per target traversal using a proposer, while exact state rebasing preserves the target trajectory. Cache capacity and multi-token acceptance are reported as different mechanisms even when they share storage and state infrastructure.

## 4 Implementation

The prototype contains a C++ Plan core, a bounded source service, coordination for host windows and GPU slots, claim-level telemetry, and model-specific adapters. Qwen3-Coder AWQ campaigns use a pinned vLLM-derived runtime; the Qwen3.6, Gemma, and Qwen3-Next campaigns use pinned llama.cpp-derived lines [9]. Each campaign has a separate source and runtime identity. The paper does not imply that results obtained in one runtime were reproduced in the other.

**Object discovery.** Adapters inventory exact native tensor extents at load time. Qwen AWQ, Qwen3.6 Q4\_K\_M, Gemma QAT Q4\_0, and Qwen3-Next Q4\_K\_M use different naming and bundle rules. Logical and aligned physical bytes are recorded separately, so padding does not inflate the capacity numerator.

**Scheduling and copies.** Layer entry requests a bundle. Misses are submitted to the source service, completed windows are copied into reserved slots, and a device event publishes readiness. Layer exit releases the consumer ticket. Host-window count and GPU-slot capacity are independent; the Qwen parity diagnostic uses 144 windows, eight slots, and 3,384 scheduled objects per request.

**Temporal state.** The Qwen3.6 path couples one target with its linear MTP block and seven physical branch states. Winner commitment and six loser rebases preserve shared hybrid state. The oracle compares every recorded target logit row, all committed tokens, shared-prefix hashes, and state equality after each commit.

**Telemetry.** The service reports submissions and completions, in-flight depth, logical and physical source bytes, padding, H2D bytes, cache events, generations, and lifecycle rejections. The outer harness records cgroup mem-

bership and events, whole-board memory and compute-process ownership, output digests, and phase times. A successful process exit alone is never a claim predicate.

**Artifact.** An audit and replay artifact accompanies this report. It preserves exact claim-bearing experimental source snapshots, reconstructs each patched runtime from a public upstream commit and patch series, and keeps portability helpers separate from the recorded experiment files. Audit verifies C1–C12 record identities and enumerated assertions offline; replay recomputes arithmetic and categorical summaries from packaged compact operands. Neither operation rederives every certificate from raw execution output. The principal C1–C3 source and execution contracts specify inputs for new executions, each with its own identity. Model weights and bulk binary outputs are not redistributed; pinned acquisition metadata, byte lengths, expected hashes, compact observations, and validators are supplied. The stable interface is `./artifact verify -claim C#` and `./artifact replay -claim C#`.

## 5 Evaluation

The evaluation asks five questions:

1. Can native execution remain full-output exact when the workload’s semantic-demand lower bound exceeds both the declared full-residency envelope and the host-hard/device-physical envelope?
2. Relative to the tested buffered path, does direct ownership change the prespecified positive-margin/no-limit-event admission outcome?
3. Against a blocking one-window direct reader, how does the complete eight-window asynchronous component change wall time at identical physical source bytes?
4. Do named invalid source and ownership transitions fail closed through two real adapter/GPU paths?
5. Does the source boundary transfer to a second adapter, and does resource admissibility imply product superiority?

Table 3 reports the evidence populations separately. Each row has its own model, runtime, workload, oracle, resource contract, cache protocol, and repetition rule.

### 5.1 Methodology and Evidence Cutoff

The evidence cutoff includes the Qwen3-Next matched-source and deterministic-fault populations, the Gemma representative fault transfer, the Qwen3-Next full-output oracle confirmation, and its demand-accounting

derivation, all dated no later than 21 August 2026. Later results are outside this report and cannot support its claims.

**Principal evidence records.** The principal results are detailed in Sections 5.2–5.4. C1 is an earlier-executable  $W/E[E_{64-out}, 64]$  certificate with clean no-fault lifecycle closure. C2 is a distinct matched-source admission and complete-component population. C3 is a later-build  $L[F]$  transition validation across two adapters. These records form one evidence package but are not pooled into one executable certificate. Temporal, Qwen-specific, shared-Plan, and Gemma cache evidence appears in Appendix A at its own identities.

**Machines.** Local RTX 3090 campaigns ran on a workstation with an AMD Ryzen 7 5800X (8 cores/16 threads), a 2 TB WD\_BLACK SN850X NVMe device, a physical 24 GiB RTX 3090, and NVIDIA driver 580.159.03. RTX 3080 Ti campaigns ran on one physical 12 GiB board with driver 580.178.04; the measurement record identifies the source filesystem as `/dev/vda1` and the backing leaf device as `vda`. The remote measurement record does not identify the physical CPU model or the provider’s underlying storage medium. Accordingly, the report makes no CPU-normalized or physical-storage-normalized performance claim for the RTX 3080 Ti rows. A public replay must capture those two identities.

**Resource observation.** Host measurements use `cgroup memory.max`, `memory.peak`, `swap`, and event counters over the complete process hierarchy. Memory peak is a kernel high-water mark rather than a sampled estimate. Whole-board GPU and foreign-process observations use 100 ms sampling for Qwen AWQ campaigns and 50 ms sampling for the Gemma and Qwen3-Next campaigns. The declared GPU budget is compared with the maximum whole-board observation; no experiment uses a hard per-process GPU partition.

**Cache and storage state.** Qwen natural utility compares the unchanged protected-LRU64 plus two-bank next-layer-prefetch mechanism with its established control; it is not a Plan campaign. Each task process contains primary and immediate-warm cycles. The separate Plan parity process also contains primary and warm requests with reset between them. Gemma cache pairs start with a cold prefill and compare zero reuse with independent per-layer LRU32, using prefill-tail seeding and alternating fixed order across five prompts. The bounded-source pair holds the Gemma capacity-32 path fixed and substitutes eight aligned `O_DIRECT` windows. Qwen3-Next uses per-layer LRU64; its long prefill warms the subsequent short decode. The later matched-source experiment holds LRU64 fixed across P, D, and F; fault cells disable cache and use eight fixed `O_DIRECT` win-

Table 3: Principal results by evidence identity. “Host-hard/GPU-audited” does not imply a hard GPU partition; the physical board is a separate capacity ceiling. Performance values are not pooled across rows.

| Campaign / system                                                             | Resource authority                                             | Exactness scope                                                                                                                | Principal result                                                                                                                                          |
|-------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Demanded weights <sup>C1</sup>                                                |                                                                |                                                                                                                                |                                                                                                                                                           |
| Qwen3-Next Q4_K_M; RTX 3090                                                   | Host hard 11 GiB; GPU audited 23 GiB on physical 24 GiB board  | $E_{64-out}$ : all 64 complete logit rows, tokens, response bytes, routes, and identities exact versus zero-cache              | $T_{repr} = 45.08$ GiB; $U_{sem}^{LB} = 43.59375$ GiB exceeds the declared 34 GiB envelope by 9.59375 GiB; 32K+64 completes with zero resource events.    |
| Matched source <sup>C2</sup> same Qwen3-Next/device                           |                                                                |                                                                                                                                |                                                                                                                                                           |
|                                                                               | Host hard 11 GiB; GPU audited 23 GiB                           | Six exact, resource-clean D/F pairs; P exact but at the host limit with 33,481 limit events                                    | Complete asynchronous-component F/D wall ratio 0.3229; all six ratios lie in [0.3211, 0.3283]; physical bytes/output ratio 1.0; churn predicate fails.    |
| Integrated faults <sup>C3</sup> later Qwen3-Next and Gemma 4 builds; RTX 3090 |                                                                |                                                                                                                                |                                                                                                                                                           |
|                                                                               | Qwen hard host 11 GiB; Gemma hard host 8 GiB; zero swap/events | All fourteen prespecified cells satisfy their validators; fatal cells publish nothing invalid; recoverable cells control-exact | Fail-closed behavior for named transitions across two real adapters in the tested binaries; not attribution to the earlier C1 executable or race freedom. |
| Bounded source <sup>C8</sup> same Gemma/device                                |                                                                |                                                                                                                                |                                                                                                                                                           |
|                                                                               | Hard host 8 GiB; 12 GiB board observed                         | Full logits, tokens, routes, and cache state exact in one pair                                                                 | Host peak 6.19 to 2.78 GiB; 14,396 balanced direct-I/O submissions and completions.                                                                       |

Table 4: Executable identity boundary for the principal results. The Plan state-machine source is byte-identical across these rows (SHA-256 prefix 131c5d0036e0); adapter and bounded-source identities are not. A passing row therefore does not transfer untested behavior to another binary.

| Record | Workload / role                  | Executable SHA-256 | Adapter / source SHA-256 | Direct attribution                                              |
|--------|----------------------------------|--------------------|--------------------------|-----------------------------------------------------------------|
| C1     | Qwen 32K+64 principal experiment | 4c68fc99e72c       | ca637a7c84e37a2d224375a5 | / Full-output exactness, resources, and clean no-fault closure. |
| C2     | Qwen 8K+128 matched P/D/F        | 7bd52530b6ad       | b126d4b58827a5f1453eb07e | / Source admission and six-pair complete-component result.      |
| C3-Q   | Qwen 512+8 named transitions     | 5083ccd5b1fa       | 029ed3af631d1c74a7cfba92 | / Ten later-build control/fault cells.                          |
| C3-G   | Gemma named-transition transfer  | a01ea9314823       | 88570b5db5a71c74a7cfba92 | / Four later-build control/fault cells.                         |

dows. Gemma fault cells use the same zero-cache, eight-window source boundary. MTP uses two 256 MiB service slots and greedy target/proposer sampling.

Repetitions. The Qwen natural population has eight control and eight candidate processes; the two request cycles within a process are not independent repetitions. The Plan comparison has 16 valid cells and was not independently repeated. The Plan parity diagnostic is one process. Gemma cache has one observation per arm and prompt; bounded source is one matched pair. The Qwen3-Next 32K+64 execution and the Qwen3.6 complete MTP request each have one claim-bearing observation after a smaller validation run. The full-64 confirmation uses one prespecified zero-cache comparator, zero statistical repetitions, and zero reruns. The matched Qwen3-Next source campaign has six preordered counterbalanced D/F pairs and a fixed paired-log  $t$  interval; P descendants stop after its first valid positive-margin/no-limit-event admission failure. That interval quan-

tifies repetition at one fixed model, prompt, horizon, runtime, device, and storage identity. It is not an interval over prompts, models, hardware, or deployments. All fourteen prespecified deterministic fault cells are one outcome-blind execution per prespecified mode, not statistical reliability repetitions. Other timing values without an independent repetition are descriptive and have no variance estimate.

## 5.2 Output-Exact Execution Beyond the Full-Residency Envelope

The principal experiment uses Qwen3-Next-80B-A3B-Instruct in native Q4\_K\_M [14, 15]. Its complete canonical tensor representation is  $T_{repr} = 48,405,005,312$  bytes (45.08 GiB). The process tree has a hard  $H_{cap} = 11$  GiB cgroup limit. The GPU contract declares and audits  $G_{decl} = 23$  GiB of whole-board use on a physical  $G_{phys} = 24$  GiB RTX 3090. Hence

Table 5: Principal C1 certificate instance. C3 is a separate later-build transition record and is not a conjunct of C1.

| Field      | C1 instantiation                                                                                                             |
|------------|------------------------------------------------------------------------------------------------------------------------------|
| <i>I</i>   | Qwen3-Next Q4_K_M, 32K+64, executable 4c68fc99e72c; 45.08 GiB representation and 43.59375 GiB semantic-demand lower bound.   |
| <i>B,A</i> | Host 11 GiB hard cgroup-charge boundary; GPU 23 GiB whole-board audit on a physical 24 GiB board.                            |
| <i>E</i>   | One same-build zero-cache comparator; exact $E_{64\text{-out}}$ over 64 outputs.                                             |
| <i>L</i>   | Ordinary publication, movement, recycle, consumer release, reset/shutdown, and clean no-fault closure. Fault cells excluded. |

$$T_{\text{repr}} - (H_{\text{cap}} + G_{\text{decl}}) = 11,897,783,296 \text{ bytes.}$$

The representation therefore exceeds the declared 34 GiB full-residency envelope by 11.08 GiB. This alone would not show that the workload touches the oversized portion. The recorded prefill trace closes that alternative: the tested adapter copies each completed stock `ffn_moe_topk` tensor before any Plan scheduling or cache action. All eight chunks contain 48 layer events, and every event’s authoritative router output contains all 512 experts. All 24,576 layer–expert identities therefore have at least one routed token. For each selected expert, the native dense expert path invokes the complete gate, up, and down projections; those three canonical extents therefore enter the semantic-demand union.

The audit joins those identities to each native gate, up, and down expert extent. Its 73,728 component intervals have zero duplicates and zero overlaps under the key (model digest, source offset, logical length). The sorted extent–union digest is `bad5b23f5ac0...`; its deduplicated size is  $U_{\text{sem}}^{\text{LB}} = 46,808,432,640$  bytes (43.59375 GiB). This conservative semantic-demand lower bound excludes the remaining 1.4869 GiB of canonical tensors rather than inferring their demand. Thus

$$U_{\text{sem}}^{\text{LB}} - (H_{\text{cap}} + G_{\text{decl}}) = 10,301,210,624 \text{ bytes,}$$

$$U_{\text{sem}}^{\text{LB}} - (H_{\text{cap}} + G_{\text{phys}}) = 9,227,468,800 \text{ bytes.}$$

The semantic-demand lower bound exceeds the declared envelope by 9.59375 GiB and the host-hard/device-physical envelope by 8.59375 GiB <sup>C1</sup>. “Demanded” means selected for at least one token, not simultaneously live. Scheduler requests and candidate cumulative source traffic—392,942,788,608 logical bytes and 395,477,815,296 aligned physical bytes across rereads—are separate quantities and are not used as the semantic-demand numerator. The Linux-visible whole-

machine observation is reported in Section 2; it is not a claim about all installed host DRAM.

One prespecified zero-cache 32K+64 oracle uses the same tested executable, model, prompt, graph, greedy sampler, source path, hardware, and resource contract as the fixed LRU64 experiment. Every byte of all 64 complete 151,936-float logit rows is equal (38,895,616 bytes per arm), as are all 64 token IDs, response bytes, all 3,408 recorded route events, and the seven-field native-consumer and two-entry model-memory destination records. Their compact, sorted-key JSON SHA-256 digests are respectively `f5ae6bc54add...` and `e23ab3127932...` in both arms, and the validator also compares the records directly. The experimental harness does not capture recurrent/hybrid-state bytes, so output equality is not promoted to a direct state-byte claim. A separate complete source-path equivalence test is also bitwise exact at the 256+8 scope defined in Section 2, but it is not pooled with the cache-policy comparison.

The principal cell processes a 32,768-token mechanics prompt and generates 64 tokens, with 63 measured decode transitions, per-layer LRU64, and a warm decode following the long prefill. Prefill takes 86.355 s and decode 6.454 s, or 9.762 transitions/s. All 64 tokens execute through the native graph, source and destination counts balance, and reset/teardown complete. Complete-process-tree cgroup-charge peak is 1,496,981,504 bytes; whole-board peak is 13,218,807,808 bytes, leaving 11,477,254,144 bytes below the declared GPU budget. Swap, OOM, memory.max, and foreign-compute events are zero.

A 512+64 mechanism diagnostic decodes at 5.037 tokens/s without reuse and 11.636 tokens/s with LRU64. It explains why retention matters at this identity but is not a policy study. The principal experiment remains one model, one device, one short warm-cache observation, and a mechanics prompt composed of four repeated copies of a held-out code fixture. It establishes resource admissibility under the prespecified hard-margin contract and full-output exactness at this 64-output identity, not semantic quality, longer-horizon stability, direct recurrent-state equality, or hardware generality.

### 5.3 Matched Source Ownership and Asynchronous Execution

The matched ownership and asynchronous-component experiment holds the Qwen3-Next model bytes, graph, sampler, per-layer LRU64, 8,192-token natural prompt, 128 greedy outputs, object order, 11 GiB cgroup, and 23 GiB GPU contract fixed. P serves extents through the kernel page cache; D uses blocking `O_DIRECT`



Table 6: All six matched complete-wall pairs and their actual D/F execution order. Ratios repeat one fixed workload/system identity; they do not sample prompts or hardware.

| Block | Order | D (s)   | F (s)  | F/D      |
|-------|-------|---------|--------|----------|
| 1     | D→F   | 102.358 | 32.867 | 0.321095 |
| 2     | D→F   | 102.228 | 32.820 | 0.321052 |
| 3     | F→D   | 100.032 | 32.837 | 0.328269 |
| 4     | F→D   | 101.996 | 32.889 | 0.322449 |
| 5     | D→F   | 102.202 | 32.887 | 0.321784 |
| 6     | F→D   | 101.616 | 32.811 | 0.322888 |

and one fixed window; F uses eight registered windows, bounded asynchronous O\_DIRECT, and the complete Plan ticket, generation, consumer, reset, and shutdown path. A separate same-build synchronous-source oracle supplies all 128 full-vocabulary logit rows, tokens, pieces, routes, source identities, and final cache state.

The first valid P cell is oracle-exact but reaches the 11 GiB ceiling and advances memory.max 33,481 times. It therefore fails the prespecified positive-headroom/no-limit-event admission predicate; this does not mean that it failed to complete. Its descendants stop as specified. All six preordered D/F pairs are oracle-exact, pass that predicate, and are lifecycle-balanced. Mean complete wall is 101.739 s for D and 32.852 s for F. The paired geometric F/D wall ratio is 0.322913, with a 95% paired-log  $t$  interval of [0.320090, 0.325762]. The F/D exposed-source-wait ratio is 0.534673, decode-rate ratio is 1.541885, and physical source bytes per output are identical (ratio 1.0). Every F cell issues and completes 55,520 source operations at peak depth eight, performs zero dynamic direct allocation, and returns all windows free.

The narrow interval measures repeatability across these six executions at the fixed identity. It is not a confidence interval for generalization across prompts, models, storage devices, or hardware.

The prespecified workload-churn premise does not hold. Decode produces 6,471 aggregate generations rather than 9,216, and the weakest layer produces 61 rather than 128. Although 3,063 of 3,072 slots recycle, the conjunctive churn criterion fails. This negative is not repaired by choosing another prompt. The supported result is that the complete asynchronous component finishes in 32.3% of D’s complete wall time at this exact point, a reciprocal  $3.10\times$  result<sup>C2</sup>. F jointly changes source concurrency from depth one to depth eight, asynchronous queueing, I/O/compute overlap, and the complete lifecycle implementation; the comparison does not attribute the difference independently to Plan bookkeeping.

#### 5.4 Integrated Deterministic Faults

The Qwen3-Next fault population uses the real graph and demanded objects at 512+8, zero cache, eight O\_DIRECT windows, and a fixed-horizon greedy sampler. Its deterministic hooks live in a later fault-instrumented binary that retains the same real adapter/GPU boundary; they are adjacent evidence for the shared core and are not attributed to the earlier executable used by the 32K+64 experiment. The earlier executable directly establishes clean no-fault closure only. One control and nine deterministic hooks cover short success, EIO, out-of-order valid completions, late stale completion, wrong object, held consumer, reset with outstanding source work, shutdown with a live consumer, and a failed extent inside a bundle. All ten material jobs are valid. The five fatal cells accept zero tokens and logits, publish no invalid readiness or H2D, explicitly retain the one interrupted Plan window, and drain the source executor. The four recoverable cells retain exact tokens, logits, routes, source identities, and final cache state against control before ordinary clean shutdown.

The representative transfer repeats no-fault, short, stale, and held-consumer cells through the real Gemma 4 graph under a hard 8 GiB cgroup. All four are valid. Short and stale accept zero tokens/logits and begin no invalid H2D; held-consumer produces bit-identical logits and exact tokens, routes, materialization count, and final cache occupancy against its material control. Across all fourteen prespecified Qwen and Gemma cells, swap, max/OOM events, and foreign compute PIDs are zero. Every terminal source executor has zero active tickets and all eight windows free<sup>C3</sup>.

These deterministic cells demonstrate that named invalid transitions are rejected at the integrated adapter/GPU boundary and that declared legal reordering remains exact. They do not sample scheduler interleavings, estimate failure probability, or prove race freedom, serving isolation, or general fault tolerance.

#### 5.5 Bounded Direct I/O Establishes Host Nonresidency

A matched Gemma pair replaces the current buffered source with the eight-window bounded O\_DIRECT executor while retaining the model, graph, cache capacity, and scheduler. The managed expert inventory is  $S = 12,846,382,080$  bytes. With  $H_{\text{cap}} = 8,589,934,592$  and declared candidate reserve  $R_{\text{policy}} = 2,935,803,872$ , target capacity while preserving the reserve is  $H_{\text{cap}} - R_{\text{policy}} = 5,654,130,720$ . Equation 1 therefore holds by 7,192,251,360 bytes; this is a policy-relative nonresidency statement, not a lower bound on unavoidable runtime state.

Table 7: Prespecified fault modes mapped to implementation-independent reuse invariants. Rows group the fourteen cells by transition; they are not statistical reliability trials.

| Mode                                                | Invariant exercised                                  | Required observation                                                           | Adapter cells          |
|-----------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------|------------------------|
| No fault                                            | Ordinary publication and closure                     | Exact output; balanced tickets, windows, slots, consumers, reset, and shutdown | Qwen, Gemma            |
| Short read; EIO; failed bundle extent               | Publish only after complete valid movement           | Zero accepted outputs, readiness, or invalid H2D; executor drains              | Qwen; short also Gemma |
| Stale completion; wrong object                      | Published version denotes requested object and epoch | Mismatch rejected before visibility; zero accepted outputs                     | Qwen; stale also Gemma |
| Valid out-of-order completion                       | Legal completion order is not overconstrained        | Full control-exact output and clean closure                                    | Qwen                   |
| Held consumer                                       | No reuse before declared consumers finish            | Reuse delayed; full control-exact output and balanced release                  | Qwen, Gemma            |
| Reset with source work; shutdown with live consumer | Drain or invalidate outstanding versions             | No stale publication; balanced terminal state                                  | Qwen                   |

The bounded source reduces complete-process-tree host peak from 6,641,348,608 to 2,980,458,496 bytes (55.123%). It issues and completes exactly 14,396 reads, with peak depth eight, zero cancellation or failure, and all eight windows free at shutdown. Cumulative logical traffic is 48,160,551,152 bytes; aligned physical reads are 48,336,400,384 bytes, including 175,849,232 padding bytes. Full logits, tokens, routes, and cache state match the control. Total wall time falls from 133.645 to 62.021 s in this single pair; the physical bound and balanced ledger, not this unrepeated timing difference, are the principal result<sup>C8</sup>.

## 5.6 Resource Admissibility Is Not Product Superiority

On five software-repository tasks, the aggregate-budget Qwen3-Next path and an aligned resident Qwen3.6 path both complete all five. The resident path is  $4.965\times$  faster in total wall time and  $14.126\times$  faster to first useful action<sup>C9</sup>. These are different model paths, not a controlled architecture comparison, and the ratios do not rank model quality. They show that storage-backed sovereignty is valuable when capacity is binding, whereas a fitting resident model remains preferable when latency is the objective.

## 6 Future Work

The matched three-arm source campaign and integrated deterministic-fault campaign are complete. A minimally hooked version of the earlier C1 source would be the direct way to bind transition validation to that executable; a no-fault cross-build parity cell alone would not establish its fault behavior. That experiment would strengthen executable-identity integration but is not required for the explicitly split certificates reported here. The next certificate-level transfer question is to apply the same inventory, authority, equivalence, and lifetime fields to an external storage-offload artifact, al-

lowing “insufficient public evidence” as a valid classification rather than presuming capability absence. Guard-mutation tests and systematic schedule exploration could strengthen causal and interleaving evidence for the lifecycle protocol. A same-implementation queue-depth sweep or a simpler asynchronous direct baseline would be required before attributing the  $3.10\times$  component result to a narrower mechanism. A second aggregate-budget model or device would test breadth.

Broader temporal-amortization and Gemma populations would estimate separate mechanism generality, not alter the completed principal result. Until such evidence exists, the Qwen3.6 timing remains a one-request result and the Gemma H2D and decode directions remain descriptive matched-Plan evidence. No result here claims a novel direct-I/O primitive, universal policy superiority, transactional semantics, or general end-to-end fault containment.

## 7 Related Work

General heterogeneous inference. FlexGen schedules computation and tensors across GPU, CPU, and disk for throughput-oriented single-GPU generation [16]. DeepSpeed Inference and ZeRO-Inference stream layers from CPU or NVMe, while DeepNVMe supplies asynchronous host/device tensor I/O and GDS interfaces [2, 4, 5]. LLM in a Flash studies flash-aware execution under limited DRAM [1], and FlexInfer uses direct I/O and fixed tensor windows for on-device inference [6]. Beyond Capacity studies dual host-bypass and GPU-direct flash paths with measured component latencies and event-driven continuous-batching evaluation [11]. These systems establish storage as an inference tier. This work’s residual claim is a recorded evidence package: explicit representation and semantic-demand inventories, hard complete-process-tree host charging, audited whole-board GPU use plus a physi-

cal capacity ceiling, full-output-horizon exactness under  $E_{64\text{--out}}$  at the principal Qwen3-Next experiment, a distinct matched ownership and complete asynchronous-component comparison, and named later-build cross-adaptor fail-closed transitions.

Sparse expert loading and caching. MoE-Infinity uses sparsity-aware expert caching on personal machines [21]. SpecOffload uses speculative decoding to unlock otherwise idle GPU capacity in an offloaded engine [23]; FlashMoE studies learned cache replacement [10]; and FluxMoE decouples expert residency using stable logical addresses [13]. Recent cache-ownership work demonstrates hard cgroup capacity for expert replay and argues that kernel LRU can be competitive with user-space ownership [17]. WiSP frames MoE serving as working-set management and reports byte-identical outputs and up to  $1.95\times$  decode throughput over static offload at matched GPU memory [22]. These results prevent cgroups, caching, or stable addresses from being treated as standalone novelty.

Closest engineering disclosures. MoE-Direct combines byte-preserving NVMe reads, bounded slots, generations, pins, and scoped token identity in a public preview artifact [19]. Its source, preview binary, and strongest reported execution identities do not fully coincide, and complete RAM headroom is not reported. MESH combines `O_DIRECT/io_uring`, fixed expert slots, reader counts, and policies, but at the evidence cutoff it is an unmerged KTransformers pull request rather than mainline behavior or a peer-reviewed result [12]. Palm’s SSD path is merged public engineering code without an associated paper; its large-model path is therefore cited as an artifact, not paper evidence [18]. WISP independently combines VRAM/RAM/NVMe tiers, double-buffered asynchronous prefetch, RAM watermarks, and VRAM planning in a public engine [20]. Colibri is the closest newly audited artifact in mechanism breadth: its fast-moving multi-family engine includes RAM/NVMe/VRAM tiers, `O_DIRECT`, optional `io_uring`, bounded asynchronous loading, byte-preserving formats, token-exact harnesses, resource planning, and documented cgroup operation [7]. The WiSP preprint and WISP engine are unrelated despite the name collision.

The refreshed audit covers public records through 20 August 2026. It makes novelty based on direct I/O, asynchronous streaming, three-tier placement, exact byte formats, or cgroups alone untenable. At the audited identities, no single disclosure was found to package the complete set of principal claims as one recorded evidence package: demand and representation arithmetic with graded authorities, the six-pair byte-matched P/D/F experiment, and all fourteen prespecified real-adaptor transition cells. This is a cutoff-scoped

public-record finding, not proof that another system lacks the underlying capability. Our evaluation likewise discloses that its C1 execution and C3 transition evidence use different binaries. The evidence package, rather than mechanism composition or a single-executable conjunction, is the residual claim.

## 8 Limitations

The principal experiment covers one Qwen3-Next quantization, mechanics prompt, tested runtime, RTX 3090, and 64-output horizon. Its semantic-demand quantity is a lower bound over managed expert bundles; it does not claim that all demanded objects are simultaneously live or inventory every nonexpert tensor touched by the graph. The full-output oracle does not expose recurrent/hybrid-state bytes or establish equality to an unmodified upstream runtime. The matched source result adds six paired observations at one natural workload; it does not establish a workload population or separate concurrency, overlap, and lifecycle implementation. P is one buffered implementation, not a theorem about every bounded page-cache design.

The RTX 3080 Ti measurement record reports the GPU, driver, virtual source device, and resource observations but not the physical CPU or provider storage model. Its timings therefore cannot support a hardware-normalized storage claim.

Complete process-tree charging and whole-board observation close the largest hidden-residency channels, but GPU use is audited rather than hard-partitioned. Here process-tree charging means cgroup-attributed memory, not every host byte causally associated with execution; per-class memory.stat and separate driver-pinned attribution were not archived. This boundary does not weaken the independent semantic-demand nonresidency result against Linux-visible host capacity plus the physical board, but it limits the 11 GiB resource predicate to the declared cgroup-charge contract. Opaque firmware accounting remains bounded by the vendor interface. The Gemma representative fault campaign does not assert page-cache first-charger ownership for its nonexpert load. Deterministic integrated fault injection does not prove race freedom, stochastic reliability, or serving isolation, and the Qwen hooks use a later fault-instrumented binary rather than the earlier principal-experiment executable. Cross-adaptor evidence is limited to Qwen and Gemma. The thin MTP margins do not establish robust deployment headroom. Finally, storage-backed execution is a capacity method; it can be substantially slower than a smaller resident model.

Table 8: Closest-record audit through 20 August 2026. Entries describe the cited public identity, not latent system capability. “Partial” means that some evidence exists but not the complete certificate field used here.

| Disclosure                 | Inventory / demand                                       | Resource authority                                    | Exactness horizon                                                | Matched ownership / integrated faults                                 |
|----------------------------|----------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------|
| This work                  | Exact $T_{\text{repr}}$ and $U_{\text{sem}}^{\text{LB}}$ | Host hard; GPU audited; physical board ceiling        | All 64 complete logits, tokens, response, routes, and identities | Six-pair P/D/F; fourteen named later-build cells across two adapters  |
| WiSP [22]                  | Working-set model; partial certificate mapping           | Matched GPU memory; partial aggregate authority       | Byte-identical outputs at reported scope                         | No corresponding fixed-identity pair in the audited paper             |
| WISP [20]                  | VRAM/RAM/NVMe planning; partial                          | RAM watermark and VRAM planning; partial              | Verified operating point; no immutable paper horizon             | No corresponding fixed-identity campaign in the audited artifact      |
| Colibri [7]                | Multi-family byte-preserving formats; partial            | Resource planning and cgroup guidance; partial        | Token-exact harnesses; partial                                   | No corresponding fixed-identity campaign in the audited revision      |
| MoE-Direct / MESH [19, 12] | Extents, slots, and versioning; partial                  | Incomplete coincident authority at audited identities | Scoped token or implementation evidence                          | No corresponding cross-adapter fault population in audited identities |

## 9 Conclusion

Memory capacity is a credible systems claim only when its budget, canonical bytes, execution oracle, and asynchronous lifetimes are explicit. The architecture separates storage ownership, physical residency, stable execution addresses, and scheduling. Its C1 certificate binds a 45.08 GiB native representation, a 43.59375 GiB semantic-demand lower bound beyond the declared and host-hard/device-physical envelopes, full-output-horizon exactness under  $E_{64-\text{out}}$ , hard complete-process-tree cgroup-charge accounting, audited whole-board GPU margin, and clean no-fault closure. Together these establish resource admissibility under the prespecified hard-margin contract<sup>C1</sup>.

Relative to the tested buffered P path, direct source ownership changes the prespecified admission outcome from exact-but-at-limit to exact with positive margin and zero limit events. Six matched direct-source pairs show that the complete eight-window asynchronous component finishes in 32.3% of the blocking one-window reader’s complete wall time at identical physical bytes per output<sup>C2</sup>. In separate later fault-instrumented binaries, all fourteen prespecified deterministic cells support fail-closed behavior for named transitions across real Qwen3-Next and Gemma 4 adapter/GPU paths<sup>C3</sup>. The appendix preserves exact temporal amortization, Qwen-specific prefetch, the shared-Plan comparison and its prespecified F1 miss, Gemma transfer, and stopped lines at their original evidence identities.

The failed churn premise, joint concurrency/bookkeeping contrast, deterministic fault schedule, audited rather than hard-partitioned GPU budget, and narrow model/hardware identities bound the claim. They forbid independent Plan causality, race-freedom, serving, or general fault-tolerance language. A fault-instrumented build from the earlier source, a broader model/device principal experiment,

an independent upstream oracle, and broader schedule exploration remain legitimate ways to increase identity integration, breadth, or independence; none is silently promoted into the scoped result.

## A Additional Experiments and Transfer Evidence

Scope. The next five subsections report earlier and cross-model experiments related to the principal mechanisms. Each retains its own oracle, workload, runtime, resource contract, and repetition design.

### A.1 Temporal Amortization with Exact Hybrid-State Rebasing

The temporal experiment uses a canonical Qwen3.6-27B Q4\_K\_M target and its linear MTP block in one composite GGUF [15]. The natural retrieval request has 8,192 prompt tokens and 512 generated outputs. Greedy target and proposer execution uses seven lanes, a maximum draft length of six, Q4\_0 KV, and two 256 MiB Plan service slots. The candidate request excludes oracle time from its wall clock.

The request completes in 42.611 s, including 7.608 s to first token, for 12.016 end-to-end output tokens/s and 14.599 generation-phase tokens/s. It requires 104 target traversals, accepts 407 draft tokens, and fully accepts 51 draft groups. After the prefill-produced first output, the 511 decode outputs give 4.913 committed outputs per target traversal<sup>C4</sup>. The state ledger records 104 winner commits, 624 loser rebases, and 728 state stages. The weight service balances 4,032 mappings and unmaps, 4,030 recycles, 26,712 consumer completions, and one shutdown.

The same-build oracle checks 723 complete logit rows, every output token, state equality after every commit, winner/loser rebasing, and shared-prefix im-

Table 9: Additional experiment populations. Values remain scoped to their own identities and are not pooled with the principal Qwen3-Next experiment.

| Campaign                              |          | Resource authority                    | Correctness or utility scope                                                              | Result                                                                                                                                       |
|---------------------------------------|----------|---------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Qwen3.6 MTP <sup>C4</sup>             | temporal | Host 7.5 GiB; GPU 11.5 GiB            | 723 full-logit rows, 512 tokens, and exposed hybrid state exact at one 8,192+512 identity | One prefill-produced output plus 511 decode outputs in 104 target traversals; 4.913 traversed outputs/traversal; 12.016 end-to-end tokens/s. |
| Qwen-specific mechanism <sup>C5</sup> |          | Host hard 6 GiB; 12 GiB board audited | Eight LongBench-derived tasks; semantic F1; 15/16 request outputs token-identical         | Equal aggregate F1 0.590909; Qwen-specific prefetch cycle 331.884 to 235.917 s (−28.916%).                                                   |
| Shared-Plan <sup>C12</sup>            | parity   | Host hard 6 GiB; board audited        | One 8,192+32 primary/warm process; token digests and lifecycle                            | Exact parity and balanced 144-window/eight-slot teardown; no performance claim.                                                              |
| Shared-Plan comparison <sup>C6</sup>  |          | Host hard 6 GiB; board audited        | Sixteen valid cells; prespecified F1 criterion fails                                      | Request+reset 335.400 to 250.785 s (−25.228%); aggregate F1 delta −0.0015625 remains a failure.                                              |
| Gemma cache transfer <sup>C7</sup>    |          | Host hard 8 GiB; 12 GiB board audited | Full logits and tokens exact in five same-scheduler pairs                                 | H2D bytes −65.021%; process physical reads −61.474%; decode wall −77.517%.                                                                   |

mutability. All pass bitwise at the scoped identity. Complete-process-tree host peak is 8,050,614,272 bytes under 8,053,063,680; GPU peak is 12,335,448,064 bytes under the declared 12,348,030,976-byte ceiling. Both margins are positive but narrow (2.45 MB host and 12.58 MB GPU). This is one developed prompt and one microbatch-dependent floating-point trajectory. It establishes temporal amortization at that identity, not a cross-prompt MTP law or a comparison with an external 8.803 tokens/s point.

#### A.2 Qwen-Specific Natural Utility Before Plan Extraction

The first Qwen AWQ population evaluates the unchanged protected-LRU64 plus two-bank next-layer-prefetch mechanism on eight support-preserving, output-blind LongBench-derived QA records [3]. It uses 8,192 input tokens, at most 128 outputs, and a hard 6 GiB complete-process-tree host boundary. This is a Qwen-specific mechanism result, not shared-Plan evidence.

Across eight control and eight candidate processes, both arms obtain aggregate F1 0.590909 and all eight task-level F1 values agree. Fifteen of sixteen request outputs are token-identical; the differing NarrativeQA warm output scores zero in both arms. Request-plus-reset cycles fall from 331.884 to 235.917 s, a 28.916% reduction, with all eight task directions favorable<sup>C5</sup>. The mechanism’s output work differs by four tokens across the population and is part of delivered utility. The supported claim is equal aggregate F1 and favorable observed cycle time on this population, not exact-trace causality or a Plan speedup.

#### A.3 Shared-Plan Parity Diagnostic

Extraction into the shared Plan is tested in a separate deterministic 8,192+32 process. It schedules 3,384 objects per request through 144 host windows and eight

GPU slots, then performs reset, an immediate-warm request, reset, and shutdown. Primary and warm token digests equal the inherited same-build oracle; requests, resets, windows, slots, consumers, and tickets are balanced at teardown. Host peak is 5,013,585,920 bytes under 6 GiB. Whole-board peak is 12,294,094,848 bytes on the 12 GiB RTX 3080 Ti, leaving the declared 590,807,040-byte margin.

This one process is implementation-parity evidence. It is not a natural-task population, multi-cell utility comparison, or a performance result<sup>C12</sup>.

#### A.4 Qwen Plan Comparison and Prespecified F1 Miss

This comparison is a third Qwen identity: 16 complete valid cells on the same model family and physical RTX 3080 Ti. The conservative nonresident qweight population is 7,247,757,312 bytes, 805,306,368 bytes larger than the hard 6 GiB host cap. Complete resource ownership, scoped Plan prefill/migration ownership, reset, lifecycle, and teardown pass.

The strongest control satisfying the exactness contract requires 335.400 s of aggregate request-plus-reset time; the Plan candidate requires 250.785 s, a 25.228% reduction. This value is distinct from the 28.916% Qwen-specific natural mechanism result<sup>C6</sup>. The prespecified official-LongBench zero-margin F1 criterion is not met: one task changes the eight-direction aggregate by −0.0015625. The candidate answer contains the exact reference “Eindhoven,” equals a valid control answer, and lies within larger cold/warm control variation. No deterministic Plan-specific error was identified, but the prespecified utility criterion failed and the cause remains unresolved. The numerical miss is not reclassified as a pass.

#### A.5 Gemma Transfer and Cache Mechanism

Gemma 4’s official QAT Q4\_0 representation supplies the second model-family path [8]. Five prompt

Table 10: Compact negative results. Each stop is scoped to its prespecified identity and acceptance criterion.

| Line                        | Result and consequence                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPT-OSS-20B contiguous pack | A local diagnostic removed 82.451% of its direct-service wall, but the physical RTX 3080 Ti cell required 22.146 s, removed only 4.014% of the same direct-service wall target, regressed its control, and reached the 6 GiB host cap with 1,349 memory.max events. The observed 1.085 GB/s carrier would need at least 5.207 GB/s. The carrier premise did not advance to the next experiment <sup>C10</sup> . |
| Llama-3.3-70B IQ2_M         | The published 24,119,299,040-byte artifact was reconstructed exactly. CPU placement of token embedding and all 80 GQA-V projections produced a 23,432 MiB GPU peak with 685 MiB margin, but only one of four untouched tasks passed. The 4/4 admission criterion stopped the planned 8,192+512 economics cell <sup>C11</sup> .                                                                                  |

pairs compare zero reuse with independent per-layer LRU32 under a hard 8 GiB cgroup and whole-board observation of the 12 GiB RTX 3080 Ti. Full logits and tokens match within all five same-scheduler, same-representation pairs. This does not assert identity to ordinary unsplit llama.cpp.

Across 501 generated tokens, H2D traffic falls from 462,068,305,440 to 161,626,889,956 bytes, a 65.021% reduction. Process block reads fall 61.474%; because page cache, readahead, and amplification mediate this counter, it is not a universal disk-bandwidth claim. Aggregate decode time falls from 1,199.513 to 269.682 s, or 77.517%, with all five directions favorable<sup>C7</sup>. One observation per cell and alternating order reduce neither storage-state nor timing uncertainty to a confidence interval.

Worst complete-process-tree host peak is 6,730,268,672 bytes, leaving 1,859,665,920 bytes. Worst whole-board peak is 6,065,553,408 bytes. Swap and resource events are zero. A same-build partial-offload comparator completes generation but reaches the 8 GiB host boundary, with 10,325 memory.max events under mmap and 7,922 without mmap; it fails the same positive-headroom/no-limit-event admission predicate under this exact host contract. Only three prompts complete their task and two reach natural EOS, so the population supports mechanism transfer and traffic reduction rather than general semantic quality.

## B Negative Results and Stopped Lines

The GPT-OSS result does not show that packing fails on a faster carrier. The Llama result does not establish a general quantization-quality law. Their role is procedural: resource or component success is insufficient when the next prospective criterion fails.

## C Paper-to-Artifact Claim Ledger

Every superscript C# resolves to one entry in claims/claim-ledger.json. In the accompanying artifact, ./artifact verify -claim C# verifies source digests, locators, and expected assertions; ./artifact replay -claim C# recomputes the packaged summary where replay is applicable. Each entry contains the paper statement, immutable result path, full SHA-256 digest, JSON pointers or a text locator, and both commands. Table 11 abbreviates the digests for print; the machine-readable ledger preserves all 64 hexadecimal digits. Artifact release v0.1.0: <https://github.com/gustavgaugue/memory-sovereign-inference-artifact/releases/tag/v0.1.0>.

## D Exact Claim Identities

The following SHA-256 values are the exact model or model-inventory identities used by the claim-bearing campaigns. Line breaks are typographic only.

Qwen3.6 target model

f0863123e9f79c1f43bb6d4f2d7bc645

41a4e92f7242c3ddae649731c03cdd09

Qwen3.6 target+MTP composite

79e80b7a2237412aad13bdd40151ab0c

04a104429182f9b719f0bb47d181b7cd

Qwen3-Coder AWQ checkpoint digest list

9da173bfd3be51db5c781dddcdf3d05b

7fa45ddd08abe7f0dc323c498fe5f218

Qwen3-Coder prepacked store

8a89a8890d2e9d08bf7d74d871b1ccc4

a7a19d74ceb944b00b237d0939dfc11

Gemma 4 QAT Q4\_0 model

3eca3b8f6d7baf218a7dd6bba5fb59a5

6ee25fe2d567b6f5f589b4f697eca51d

Qwen3-Next Q4\_K\_M model file

d103b2733ec1012a52d01edda66b7e5c

24ae50508c9f99f5297ea459ef3c061a

Runtime commits.

Qwen3.6 base

1dfb398ea4317c9204e40aa860c0c251ab74d8a4

Qwen AWQ

b29d00067fe3d641ba2995b7c693d8bea8c83a8f

Gemma base / adapter

40d5358d3c730b81729ba81cd5c44ed596d02510

05f332a0b031aa18041622e26f37e6a7c1fe82f0

Qwen3-Next base / adapter

8ef78e644f559db4e8716b59bf76b8e11619337d

7206605099424f7ad7ca78c2783a60ec60f810f4

Table 11: Mechanical paper-to-artifact claim map. The stable artifact commands substitute the listed ID for C#.

| ID  | Headline family                               | Immutable compact result | SHA-256 prefix |
|-----|-----------------------------------------------|--------------------------|----------------|
| C1  | Semantic demand beyond full residency         | Publication claim record | af894187b754   |
| C2  | Six-pair complete asynchronous component      | Publication claim record | 29ebe2b94cf3   |
| C3  | All fourteen prespecified fault/control cells | Publication claim record | c2ca7498e92c   |
| C4  | Qwen3.6 MTP temporal amortization             | Publication claim record | 3aa1cd4f8545   |
| C5  | Qwen-specific natural utility                 | Publication claim record | 67922edcd38e   |
| C6  | Shared-Plan comparison and preserved F1 miss  | Publication claim record | 72b59f174b88   |
| C7  | Gemma cache transfer                          | Publication claim record | ef3215cfd764   |
| C8  | Gemma bounded direct source                   | Publication claim record | 4432f76df666   |
| C9  | Cross-model product comparison                | Publication claim record | b3a576f4c5a8   |
| C10 | GPT-OSS storage-device stop                   | Publication claim record | 03a01c6ededb   |
| C11 | Llama task-admission stop                     | Publication claim record | cc4865e51d59   |
| C12 | Shared-Plan parity process                    | Publication claim record | 88e7fa900277   |

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